HCL | CISCO XDR PROOF OF VALUE

Complete Build and Recreation Report

What was built, how it maps to the deliverables sheet, where each item stands, and how to recreate it

Prepared by	Dev Bhargav
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Executive summary

Purpose. This report documents the Cisco XDR proof of value (PoV) built for HCL. It records what was built, how each piece maps to the deliverables sheet, where each item stands today, and the steps needed to recreate the environment.

Outcome to date.

- **Proven end to end:** workflows started by an automation rule, host containment through Cisco Secure Endpoint on a lab machine, and Webex notification of new incidents.
- **Built, awaiting confirmation:** the custom playbook with its assignment rule, ServiceNow ticketing, and the AWS integration, which is not yet showing data in XDR.
- **In progress or not yet working:** the AI agent (in progress), Jira (unconfirmed) and email notification (blocked by SMTP sign-in). Webex covers the notification need in the meantime.

The full status table is in section 1.

Risks and dependencies

Item	Effect	Action needed
AWS data not yet visible in XDR	The public cloud integration row cannot be marked Met	Re-add the credential with the correct ARN, check the S3 files and the AssumeRole event, and confirm region support (`ap-south-2`) with Cisco TAC
Email notification blocked	The email part of the trigger row is incomplete	Provide an SMTP account that allows sign-in, or accept Webex as the notification route
Endpoint product	The sheet names CrowdStrike; this build uses Cisco Secure Endpoint	HCL to confirm which product is expected
AI assistant availability	One AI row is marked Roadmap in the sheet	Record in the Notes column whether the feature was available in the tenant
Evidence not yet captured	No sheet row can be marked Met without it	Collect the evidence listed in section 13

Decisions requested

1. Confirm whether Cisco Secure Endpoint or CrowdStrike is the expected endpoint product, and what "ERD-M" refers to.
2. Agree the Level of Importance value for each row of the deliverables sheet.
3. Confirm ServiceNow as the ticketing route, and either fix or retire Jira.
4. Provide an SMTP account that allows sign-in, or accept Webex as the notification route.

Next steps

1. Confirm that AWS data appears in XDR.
2. Test the ServiceNow create-or-update workflow and add it to the playbook.
3. Add outputs to the containment workflow, add the containment task to the playbook and walk a new incident through it.
4. Finish and test the AI agent.
5. Capture the evidence and update the deliverables sheet.

1. Purpose and how to use this report

This report records everything built in the HCL Cisco XDR proof of value (PoV), in the order it was built, so you can rebuild it alone and map each piece to a row of the deliverables sheet.

1. Read section 2 and gather every account and credential it lists.
2. Look at the diagram in section 3 to see how the parts connect.
3. Use section 4 to see which sheet row each piece proves and where it stands.
4. Build in order with sections 5 to 10. Each one lists the clicks, the values and the traps.
5. Prove each piece with section 11, and use section 12 when something fails.
6. Collect the evidence in section 13 before changing a row to "Met" in the sheet.

Status words used throughout. Proven: it ran and you saw the result. **Built**: configured, not yet confirmed. **Partly done**: some of the row is covered. **In progress**: being worked on. **Blocked**: waiting on something outside XDR. **Unconfirmed**: tried, result not checked.

Menu names and field labels in XDR change between releases. Where a label differs on your screen, follow the description of what the step does. No passwords, tokens or keys appear here; keep them in your own password manager.

Source: the project documents in the XDR project (build guide, work summary, R.A.W build pack, AWS guides) and the working session notes.

Status at a glance

Area	Status	What is left	Section
Workflows and automation rule	Proven	Tidy the rule's title, condition and type	6.1
Custom playbook and assignment rule	Built	Walk a new incident through every task	6.2
Containment with Secure Endpoint	Proven	Map outputs, add to the playbook, capture evidence	7
Webex notification	Proven	Replace placeholder text in the message	8.1
ServiceNow ticketing	Built	Run on a test incident and check for duplicates	8.2
AWS integration	Built	Get flow log and CloudTrail data to appear	9
AI agent	In progress	Finish the form and test on an incident	10
Jira	Unconfirmed	Confirm the fix or retire it	8.3
Email	Blocked	Needs an SMTP account that allows sign-in	8.4

2. Environment and prerequisites

The work runs on one tenant, `xdr.apjc.security.cisco.com` (Asia-Pacific, Japan and China region), with an Advantage licence, and needs the accounts below before any workflow will validate.

Item	What you need	Used for	Status
Cisco XDR account	Role allowed to create workflows, rules, playbooks and integrations	Everything	Proven
Webex	A bot from developer.webex.com, its access token, and a space the bot has joined	Notifications	Proven
ServiceNow	Developer instance `dev453866.service-now.com`, user `xdr_integration` with the `itil` role	Ticketing	Built
Jira Cloud (alternative)	Site URL, account email, API token, project key, issue type	Ticketing	Unconfirmed
Email (SMTP)	SMTP host, port and a login that allows SMTP AUTH	Email notification	Blocked
Cisco Secure Endpoint	Console API credentials with Read and Write scope, a connector on a test machine, Host Isolation enabled in that machine's policy	Containment	Proven for the isolate run
AWS	Account `886571876250`, an IAM role `cisco-xdr-integration`, the XDR Organization ID as external ID, S3 buckets for logs	Cloud telemetry	Built no data yet
Test machine	A lab machine with the Secure Endpoint connector, on a separate network	Containment and detection tests	Proven

Region endpoints used for API work: `visibility.apjc.amp.cisco.com/iroh/` and `automate.apjc.security.cisco.com/api/`.

Every workflow reaches an outside system through a **target** (where to connect) and an **account key** (the stored credentials). Create the integration first, then the target, then the workflow. Without them the workflow will not validate.

3. Architecture at a glance

A high-priority incident follows two paths at once: an automation rule runs the Webex and ServiceNow workflows immediately, and an assignment rule attaches the playbook that the analyst follows. The diagrams below are Mermaid source. If one shows as plain code, open the matching image in the `diagrams` folder, or paste the source into mermaid.live.

3.1 End-to-end flow

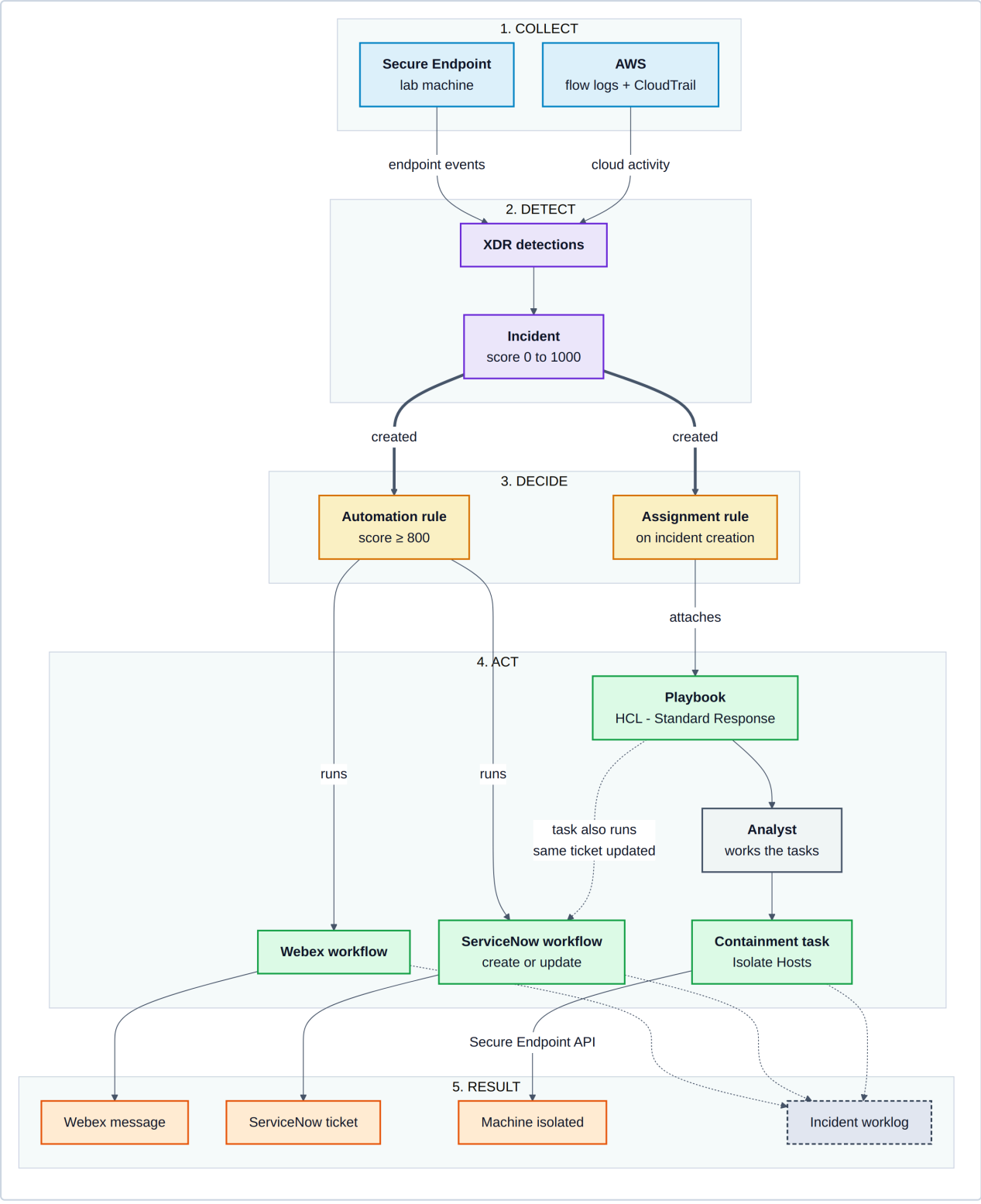


Figure 3.1: End-to-end flow. Blue is a data source, purple is XDR's own analysis, amber is a rule that decides, green is automation you built, grey is the analyst, orange is the visible result and the dashed box is the record. The thick arrows show the incident passing to both rules at the same time. The dotted arrow from the playbook shows that it can also run the ServiceNow workflow, which then updates the same ticket instead of creating a second one. The containment task acts on the same machine that produced the detection.

3.2 Containment workflow

The workflow is shown in two parts: how the machine is identified, then how it is isolated and reported.

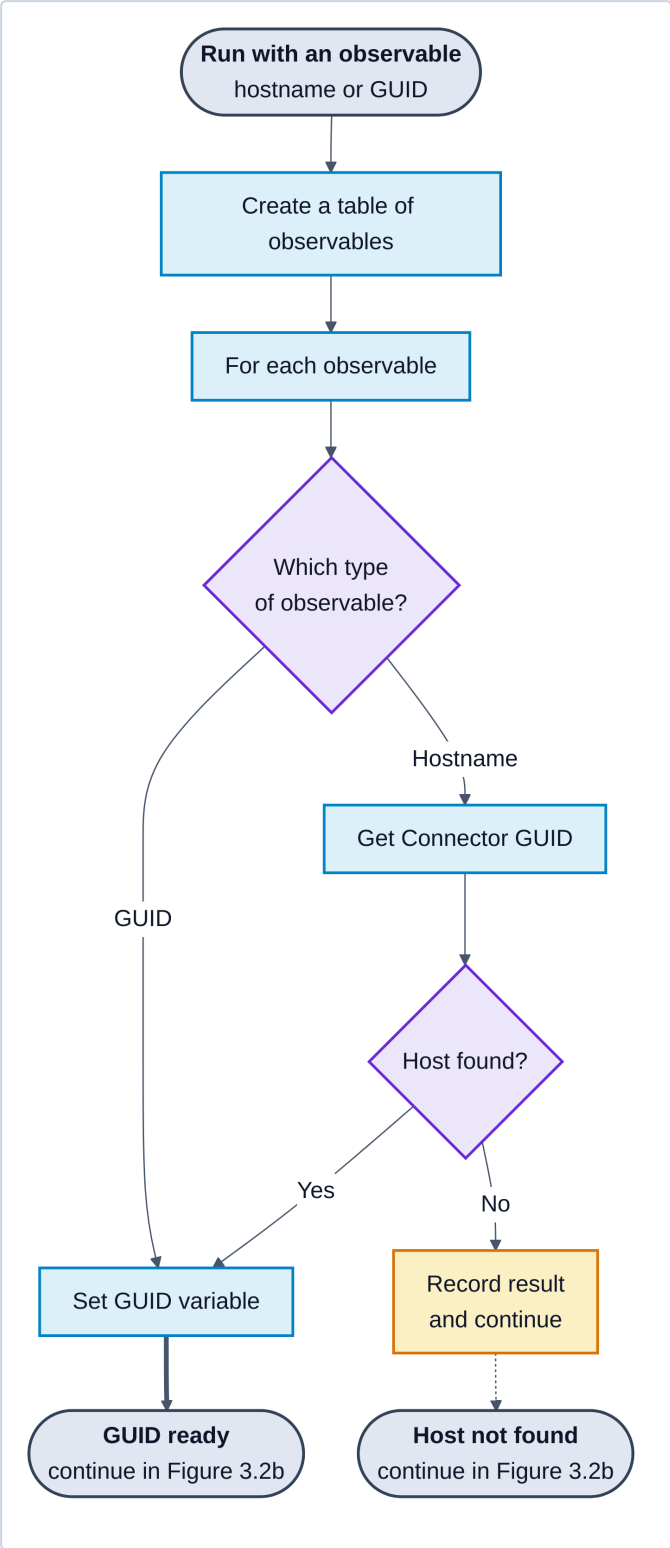


Figure 3.2a: Identify the machine. Both entry points join at the connector GUID, so a hostname and a GUID share the same lookup and isolation steps. A host that is not found is recorded and the workflow moves to the next observable.

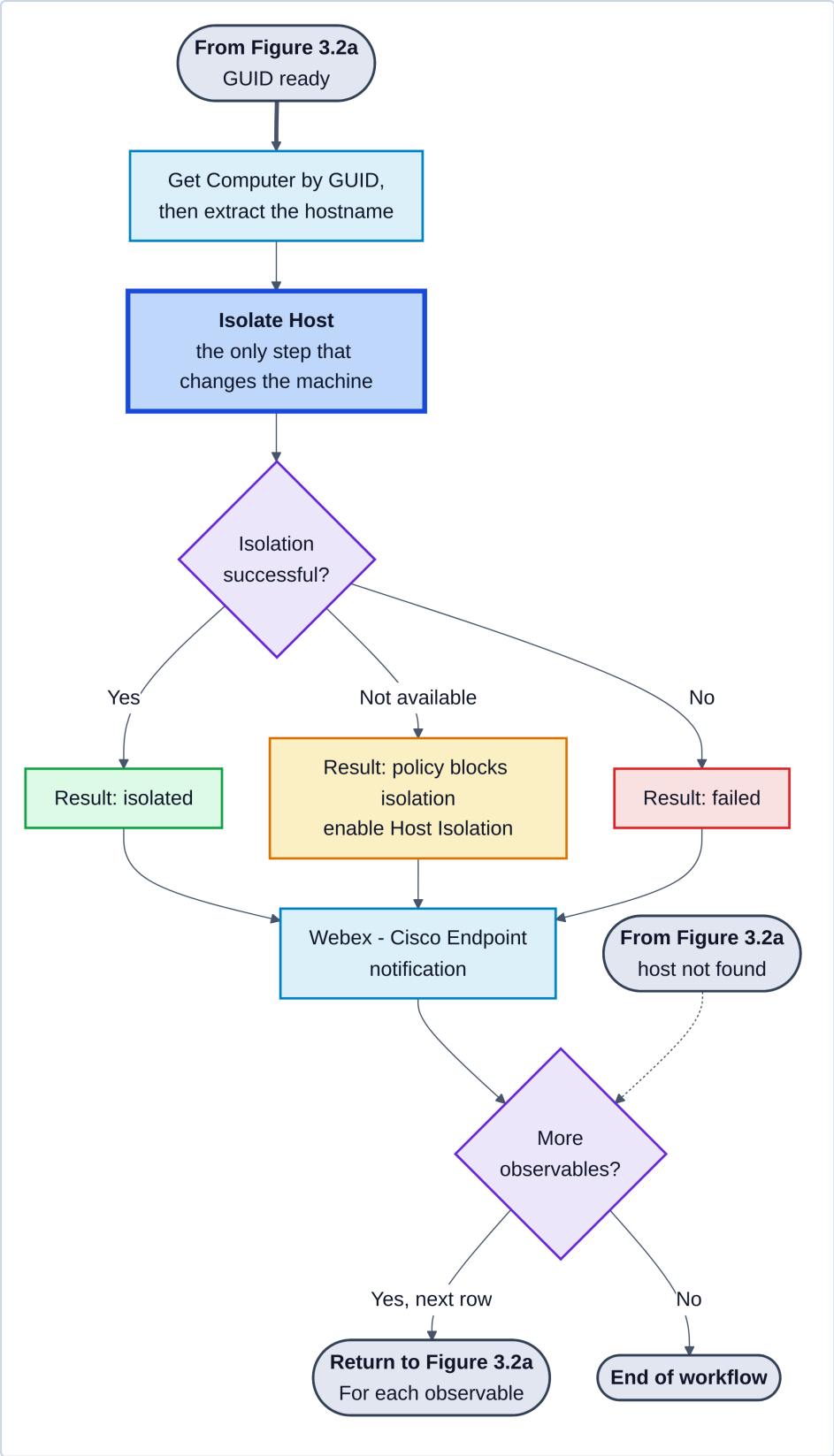


Figure 3.2b: Isolate and report. The heavy blue box is the only step that changes anything on the machine. Every path, including a host that was not found, ends at *More observables?*, so the loop always finishes and the workflow ends cleanly.

3.3 AWS data path

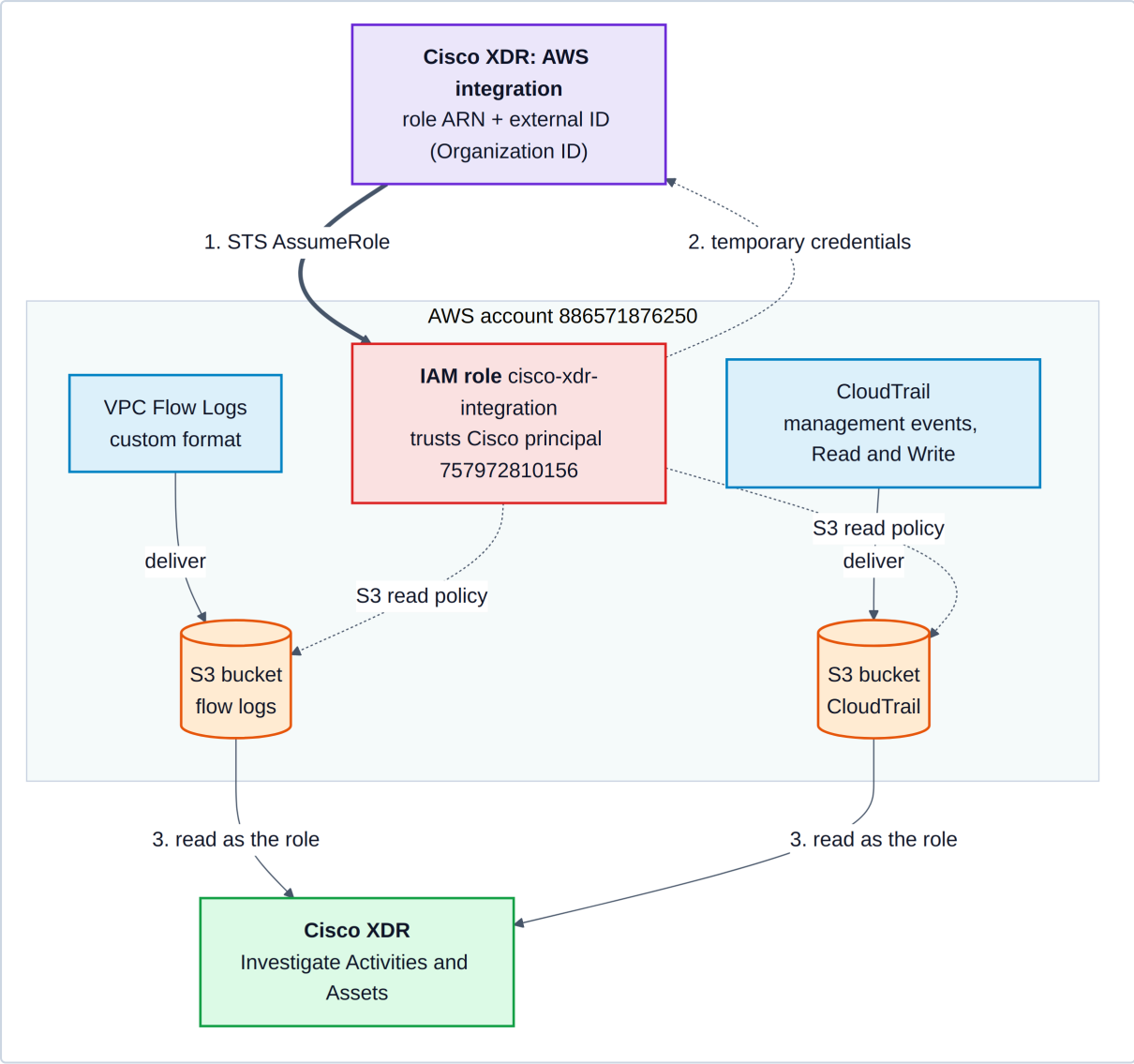


Figure 3.3: AWS data path. XDR never holds AWS keys. Step 1 (thick arrow) is where a wrong ARN or external ID stops everything, before any data is read. Once AWS returns temporary credentials (step 2), XDR reads the two buckets as the role (step 3). The dotted lines show the S3 read policy that lets the role see each bucket.

3.4 Playbook phases

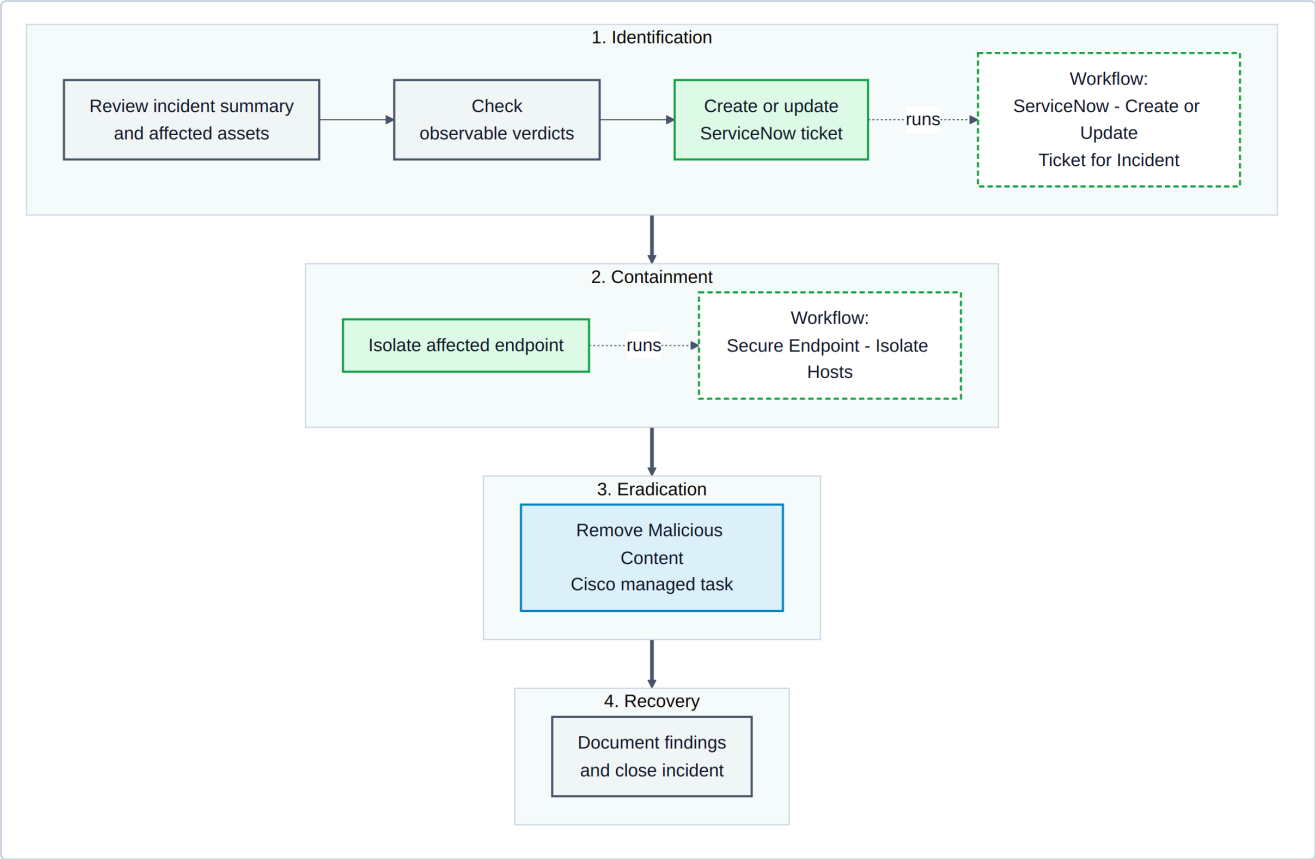


Figure 3.4: Playbook phases. Grey tasks are done by the analyst, green tasks run a workflow (shown in a dashed box) and the blue task is Cisco-managed. The phases are fixed and run in order; only the tasks inside them are yours.

3.5 Who starts what

The timeline shows one incident from creation to closure. The two rules start at the same moment; the analyst then works through the playbook.

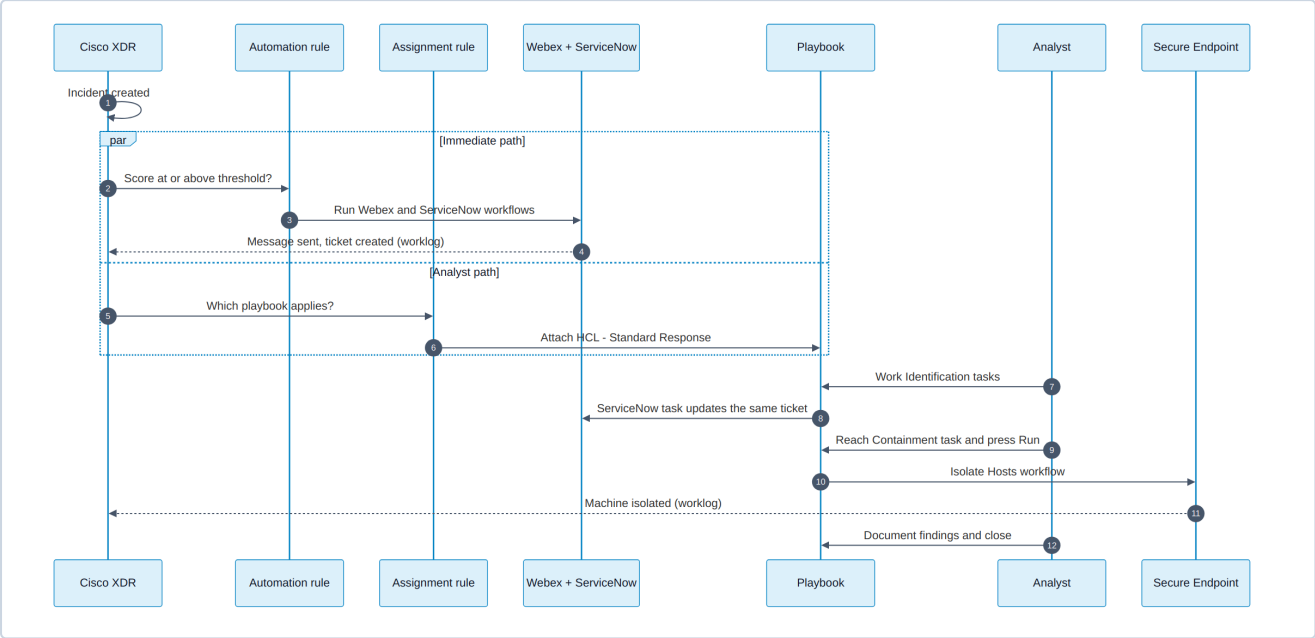


Figure 3.5: Incident timeline.

The same information as a table:

Trigger	Starts	Result
Incident created at or above the threshold	Automation rule	Webex message and ServiceNow ticket
Incident created	Assignment rule	Playbook attached to the incident
Analyst reaches the Containment task	Playbook task	Isolate Hosts workflow runs
Analyst runs it by hand with an observable	Manual run	Isolate Hosts workflow runs

4. Mapping to the deliverables sheet

The work covers the four R.A.W rows, plus parts of Integration, Incident Management and AI. Rows you did not work on are left out. Level of Importance and Requirement Met are still blank in the sheet, so agree levels with HCL before marking anything Met.

Response, Automation, Workflow (R.A.W)

All four rows target AR (Accelerate Response).

Sheet row	Status	What you did	Section
Create Ticket (ServiceNow)	Built	Created the developer instance and the `xdr_integration` user, and chose the Cisco-managed "Create or Update Ticket for Incident" workflow. The test run is still to confirm.	8.2
Complete Containment Workflow	Proven	Ran a copy of the Cisco-managed "Secure Endpoint - Isolate Hosts" workflow on the lab machine. Host isolation only.	7
Create a Trigger (priority score above 800)	Partly done	The automation rule fired and started the workflow, and the Webex message works. Email is blocked.	6.1, 8
Create a Custom Playbook	Built	Published `HCL - Standard Response` with four phases and created its assignment rule. A full walk-through is pending.	6.2, 6.3

Integration

Both rows target DS, PI, SI and AR (Detect Sooner, Prioritize by Impact, Speed up Investigation, Accelerate Response).

Sheet row	Status	What you did	Section
Integrate with Endpoint Solution	Partly done	Integrated Cisco Secure Endpoint and used it for isolation. The sheet names CrowdStrike, so confirm which product HCL expects.	7
Integrate with Public Cloud Infrastructure	Built	Added the AWS role, VPC Flow Logs and CloudTrail. Data is not yet appearing in Investigate or Assets.	9

Incident Management (I.M)

Sheet row	Status	What you did	Section
Investigate Incident	Partly done	Opened and previewed incidents; INC-18 was used for the Webex message tests.	8.1

Sheet row	Status	What you did	Section
Complete Incident Lifecycle to Closure	Partly done	Built the custom playbook. Closing an incident through every task is still to do.	6.2
Review Incident Worklog	Partly done	Run history shows the rule as the starter of each workflow run. The closed-incident worklog review is pending.	6.1

Artificial Intelligence (A.I)

Sheet row	Status	What you did	Section
Use the AI assistant to interact with an incident (marked Roadmap in the sheet)	In progress	Worked through the agent creation form. Use on an incident is not yet confirmed.	10

Supporting work

Item	Status	Used for
Webex notification	Proven	Telling the response team about a new incident
Jira Cloud	Unconfirmed	Alternative ticketing; the first run failed on two field values
Email notification	Blocked	Priority-score email; blocked by SMTP sign-in

The AWS beginner guide, the AWS setup request for the cloud team and this report are the written outputs.

5. Foundations: targets, account keys and a first workflow

Build the Hello-XDR workflow first: it needs no outside system, so only your logic can fail, and it teaches the editor and the run view.

Key terms

Term	Meaning
Workflow	A flowchart of steps that does one job
Activity	One step in the flowchart
Atomic action	A prebuilt single step for a product, such as one Secure Endpoint call
Target	A saved connection to an outside system
Account key	The stored credential a target uses
Variable	A named value: Input (passed in), Output (returned), Local or Static (inside)
Intent	The workflow type chosen at creation: Incident Response, Observable action, or general
Observable	An IP, domain, hash, hostname or similar indicator

Hello-XDR (proven)

1. Go to Automate, then Workflows, then New workflow. Name it `Hello-XDR` and use a general intent.
2. Declare four variables: `name` (input), `ip` (input), `is_private` (local), `message` (output). An undeclared variable makes validation fail.
3. Add a condition on `ip` that checks for the private ranges `10.`, `192.168.` and `172.16.` to `172.31.`.
4. On each branch set `message`, inserting values with the `(x)` variable picker, or type `$` to search by name.
5. Click Validate and fix any red errors. Then Run it with test values and open each step in the run view to read its inputs and outputs.
6. Run again with `8.8.8.8` to prove the other branch.

Variable picker

The picker has five columns of sources: Activities (outputs of earlier steps), Env, Global, Incident response and Workflow. In a step's output you can choose fields such as Output, Row Count, Start time, End time or Elapsed time. Inside an array, Items is every row, First and Last are single rows, and Columns lists the names. A greyed entry means its type does not fit the field you are editing.

Rule-Test workflow (proven)

This is the smallest useful Incident Response workflow, used to prove rules and playbooks can start work.

1. New workflow, intent **Incident Response**, name `Rule-Test`.
2. Add one **Set Variables** step that writes a short message, for example the incident title.
3. Validate, install, run once by hand, then open the run and confirm each step passed.

Two habits that save time

- Read the child step when a run fails. The parent run only says that something went wrong.
- Leave **Continue workflow execution on failure** unticked while testing. When it is ticked, a failed step looks like a success.

6. R.A.W: the automation rule, assignment rule and playbook

Two separate mechanisms respond to a new incident: an **automation rule** acts instantly, and an **assignment rule** attaches a **playbook** that an analyst follows. Build them in this order.

6.1 Automation rule (sheet row: Create a Trigger)

1. Go to Automate, then Automation Rules, then New Rule.
2. Set the trigger to **incident created** (or the status change you want).
3. Add a condition on **priority score** at or above your threshold, for example 800. The score runs from 0 to 1000.
4. Under Workflows add `Rule-Test`, then later the Webex and ServiceNow workflows.
5. Choose the rule type. **Standalone** rules run on their own. **Priority** rules are evaluated in order and can stop later rules, so use standalone unless you need that ordering.
6. Save and enable it.

Prove it fires. Set the threshold low (for example 100) so an existing or new incident matches. Open the workflow's Runs page and confirm a run with that incident ID and the rule as starter. Then raise the threshold to 800 and note which incident exceeds it.

Tidy-up still to do. Make the rule's title, condition and description agree; an earlier rule still said "Runs Rule-Test" after its condition changed. The rule was also switched from standalone to priority and should go back.

6.2 Playbook (sheet row: Create a Custom Playbook)

A playbook is the analyst's checklist, split into four fixed phases: Identification, Containment, Eradication and Recovery. You add tasks inside each phase. A task is **manual** (the analyst does it) or **automated** (it runs one workflow). Cisco-managed tasks can sit beside your own.

1. Go to Automate, then Playbooks, then New Playbook. Name it `HCL - Standard Response`.
2. Add the tasks below, then Save and Publish. Publishing locks that version; later changes need a new version.

Phase	Task	Type	Workflow
Identification	Review incident summary and affected assets	Manual	none
Identification	Check observable verdicts	Manual	none
Identification	Create or update ServiceNow ticket	Automated	ServiceNow - Create or Update Ticket for Incident
Containment	Isolate affected endpoint	Automated	Cisco Secure Endpoint - Isolate Hosts By Dev Bhargav
Eradication	Remove Malicious Content	Cisco managed	managed task
Recovery	Document findings and close incident	Manual	none

The first version also held a "Run test workflow" task using `Rule-Test` to show that tasks can run workflows. Drop it once the real tasks work.

Rules for tasks

- A task runs **one** workflow, so ticketing and containment are separate tasks.
- The workflow must be validated and, in some versions, enabled for incident response before it appears in the task picker.
- A playbook task passes the incident's observables to the workflow, so the incident needs a hostname or GUID observable for containment to act.
- Keep Webex in **either** the rule **or** the playbook. In both, the room gets two messages for one incident.
- Many teams make the isolation task manual so an analyst presses Run. That is safer; keep it automatic only if the PoV needs full automation.

6.3 Assignment rule

An assignment rule decides which playbook is attached to a new incident. Create one that attaches `HCL - Standard Response` to incidents you care about, for example by priority. **It applies only when the incident is created**, so old incidents never receive the playbook. Test with a new incident.

6.4 Suggested description text

- **Rule:** Runs for incidents with priority score above 800 to notify the response team and start automated steps.
- **Playbook:** Standard triage, containment and closure steps for the HCL PoV. Guides an analyst through identification, containment, eradication and recovery.
- **Contain task:** Isolate the affected endpoint after approval and record the result in the worklog.

6.5 Recommended wiring

Piece	Job	Starts when
Automation rule (standalone)	Webex message and ServiceNow ticket	Incident created at or above the threshold
Assignment rule	Attaches the playbook	Incident created
Playbook	Analyst checklist and containment	Attached by the assignment rule

Use the **Create or Update** ServiceNow workflow in both the rule and the playbook. If both run it, the second run updates the existing ticket and does not create a duplicate.

7. Containment with Cisco Secure Endpoint

The containment workflow isolates a machine through Cisco Secure Endpoint from an observable (hostname or connector GUID), and the isolate run on the lab machine was reported successful.

7.1 Prerequisites

1. In the Secure Endpoint console create API credentials with **Read and Write** scope. Whoever has console access does this.
2. In XDR go to Administration, then Integrations, and add Secure Endpoint with those credentials. Confirm it shows Connected and a target exists.
3. Install the Secure Endpoint connector on the lab machine. The machine can sit on a different network as long as it can reach Cisco's cloud.
4. In the policy applied to that machine, enable **Host Isolation**. Without it the workflow reaches the "Isolation not available" branch.

7.2 Build the workflow

1. Open Exchange and install the Cisco-managed **Cisco Secure Endpoint - Isolate Hosts** workflow. Copy it to your own tenant version, here named `Cisco Secure Endpoint - Isolate Hosts By Dev Bhargav`.
2. On the install page choose your Secure Endpoint target, then validate.
3. The workflow follows this path for each observable:

Step	What it does
Create a table for observables	Builds a table from the observables passed in
For each observable	Loops over each row
Which type of observable is this?	Branches on Hostname or GUID
Hostname branch	Runs Get Connector GUID, asks "Was a host found?", and sets the GUID variable on Yes or records a result and continues on No
GUID branch	Sets the GUID variable directly
Get Computer by GUID	Looks up the machine
Extract the hostname	Pulls the name from the lookup
Isolate Host	Sends the isolation request
Isolation successful?	Yes, Isolation not available, or No; each updates the workflow result
Webex - Cisco Endpoint	Child workflow that posts the outcome

7.3 Run it

1. Click Run and add an observable. Use the **hostname** (run `hostname` in a command prompt on the lab machine) or the connector **GUID**.
2. Open the run, click the For each iteration, and check three things: Get Computer by GUID returned a computer, Isolate Host returned a response, and which branch of "Isolation successful?" ran.
3. Confirm on the machine: the Secure Endpoint app shows isolation, or network access stops. It can take a minute or two.

Finding the GUID without console access. Read it in the connector app's settings or about view (look for a GUID, UUID or Connector ID field), generate the connector support package, or search the hostname in Investigate in XDR and read the endpoint's GUID from the result. The hostname route is simplest, because XDR resolves it itself.

Why the output shows `{}`. The workflow only sets variables. A run's output lists variables marked as workflow outputs, and none are, so the output is empty even on success.

7.4 Variables to add

Create these in the Variables panel and fill them with Update variable steps beside the existing result steps.

Variable	Scope	Set it from	Why
`incident_id`	Input, optional	Playbook or rule	Links the action to the incident
`hostname`	Output	Extract the hostname step	Shows in Webex and tickets
`computer_ip`	Output	Get Computer by GUID	Evidence
`os_version`, `connector_version`	Output or Local	Get Computer by GUID	Evidence
`policy_name`	Output	Get Computer by GUID	Explains "isolation not available"
`last_seen`	Output	Get Computer by GUID	Shows whether the host was online
`isolation_status`	Output	Isolate Host response	One value: isolated, not_available or failed
`error_message`	Output	Failure branches	Fast troubleshooting
`unlock_code`	Output, sensitive	Isolate Host, if returned	Needed to release the host
`timestamp`	Local	Format Date	Evidence trail

> **Handle with care.** Keep `unlock_code` out of Webex and email; put it in the ticket work notes. Field names in the picker depend on what Get Computer by GUID returns, so open a successful run and use the picker to see them.

7.5 Reverse and approvals

Build or install a stop-isolation workflow so the lab machine can be restored after each test. If an unlock code was returned, it is needed here. For production, add an approval step before Isolate Host and a protected-hosts list for domain controllers and core servers.

7.6 Attach to the playbook

In the Containment phase of `HCL - Standard Response`, add an automated task that runs this workflow (section 6.2).

7.7 What is still open

- The sheet names CrowdStrike for containment; this build uses Secure Endpoint. Confirm with HCL which one they expect.
- Capture evidence: the workflow run, the isolated machine, the console showing isolation, and the reversal.

8. Notifications and ticketing: Webex, ServiceNow, Jira, email

Webex is the only notification path proven so far; ServiceNow is the chosen ticketing route, and Jira and email remain open.

8.1 Webex (proven)

1. At developer.webex.com create a **bot** and copy its access token. Add the bot to your Webex space; it can only post where it is a member.
2. In XDR go to Administration, then Integrations, enable Webex and save the token as the account key.
3. In Automate, then Targets, check a Webex target exists. "No valid account key" means the integration is not connected yet.
4. Open Exchange, install **Webex - Send Message for New Incidents**, and fill **Sources** and **Webex Room Name**.
5. Pick your Webex target when warned. This clears "Execute on this target group" errors.
6. Validate, install, and attach the workflow to the automation rule (section 6.1).

Customising the message. Put your text in the **Markdown Message** field and leave **Plain Text Message** empty. Replace every `<<PLACEHOLDER>>` with a value from the `(x)` picker, or type `$`. Typing an incident's values by hand makes every alert show the same incident.

Placeholder	Insert from the picker
Incident ID	Incident ID or number
Title	Incident title
Priority score	Priority score
Assessment	Assessment
Sources	Sources list
Status	Incident status
Created, last updated	Created and updated time
Link	Incident link

The Webex message still contained placeholder text when last checked. Templates are in `webex_incident_message.md`.

Room not found. The error `room "XDR_Automation_Action_Notification" not found` means the room name must match an existing space exactly. Either change the room variable to the space the bot is in, or create a space with that name and add the bot.

8.2 ServiceNow (built, test pending)

1. Wake the developer instance `dev453866.service-now.com`; idle instances hibernate.

2. In User Administration create `xdr_integration`. Use Set Password, untick **Password needs reset**, tick **Internal Integration User**, set it Active, and add the **itil** role.
3. Test from a laptop with `curl -u xdr_integration:PASSWORD "https://dev453866.service-now.com/api/now/table/incident?sysparm_limit=1"`. JSON means it works. 401 is credentials; 403 is a missing role.
4. In XDR enable the ServiceNow integration (host, username, password), refresh the install page, and pick the target.
5. Install **ServiceNow - Create or Update Ticket for Incident** (v1.0). Fill Caller User ID with `xdr_integration`, and leave Assignee and Assignment Group blank because a wrong group name can reject the ticket.
6. Run it on a test incident and find the ticket under Incident, then All, in ServiceNow. Run it again: the same ticket should update and no second one appear.

The alternative workflow, "Create Ticket for Incident" (v1.1), only creates, so it can duplicate. Both set the XDR incident status to Incident Reported.

8.3 Jira Cloud (alternative, unconfirmed)

The first run failed because Project Key was `KTHHZXKY-1` and Issue Type Name was "Security Incidents". Use the key without `-1`, an issue type that exists such as Task, and clear Project ID, Issue Type ID and Watchers. Read the child "Jira Cloud - Create Issue" step for the real error. Retire Jira if ServiceNow is the chosen route.

8.4 Email (blocked)

The email workflow needs an SMTP account that accepts a password sign-in from XDR. Outlook and Microsoft 365 often block SMTP AUTH; SendGrid needs a verified single sender; a Gmail app password works for a lab. Until fixed, Webex covers the notification requirement.

9. AWS integration

The AWS integration was configured end to end, but no VPC Flow Log or CloudTrail data has appeared in Investigate or Assets yet, so treat it as Built, not Proven. XDR reads two kinds of AWS data: **VPC Flow Logs** (who talked to whom on the network) and **CloudTrail management events** (who did what through the AWS API). It needs read-only access through a cross-account IAM role.

9.1 What the cloud team must create

The request sent to the cloud team (`AWS setup request for Cisco XDR`) asks for these items. Gather them before touching XDR.

Item	Value or rule
AWS account ID	`886571876250`
Region of the workloads and buckets	Confirm the region; `ap-south-2` support is unconfirmed, so ask Cisco TAC
S3 bucket for flow logs	Created in the same region as the VPC
S3 bucket or path for CloudTrail	A trail with management events set to **Read and Write**
IAM policy	The S3 read policy that XDR's Permissions tab shows for your tenant
IAM role	`cisco-xdr-integration` , trusting Cisco principal `arn:aws:iam::757972810156:root`
External ID	Your XDR **Organization ID** (the trust condition on the role)
KMS	If the buckets are encrypted, allow the role to decrypt

> **Common trap.** The complete ARN must read `arn:aws:iam::886571876250:role/cisco-xdr-integration` . The earlier attempts failed because `:role` was missing: `arn:aws:iam::886571876250/cisco-xdr-integration` is invalid and returns a 400 from STS.

9.2 Flow logs

1. In the VPC console, create a flow log for the VPC or subnets with the destination set to your S3 bucket.
2. Choose a **custom format** that includes all version 2 attributes (version 3 recommended).
3. Include `pkt-srcaddr` and `pkt-dstaddr` so NAT gateway traffic stays visible.
4. Exclude `reject-reason` .

Flow logs only contain data when there is traffic. An idle VPC with no running instances produces nothing, which alone explains an empty Investigate page.

9.3 CloudTrail

Create a trail that logs **management events, Read and Write**, delivered to S3. Data events are not logged by default and are not needed. Per the earlier notes, XDR's default collection path for CloudTrail is the CloudTrail Lookup API rather than the S3 trail, so confirm with Cisco which path your tenant uses.

9.4 Connect it in XDR

1. Go to Administration, then Integrations, then Third-Party, and add AWS (Advantage tier).
2. Add a credential: paste the role ARN, the account ID and the region. The health check runs an STS AssumeRole call, and a failure message names the reason.
3. Open the **VPC Flow Logs** tab, add the bucket and pick the credential.
4. Add the CloudTrail S3 path if you use one.
5. Confirm the **Credentials** page shows the account as Connected.

If the credentials dropdown is blank, select the checked option and see whether Monitor status populates. If it stays blank, re-add the credential with the correct ARN or raise a Cisco case.

9.5 Verify

- **XDR:** Monitor status fills in for the flow log; flows appear in Investigate, then Activities within about 10 minutes of delivery; entity and asset data can take up to 24 hours.
- **AWS:** open CloudTrail Event history and look for an `AssumeRole` event from Cisco's principal. It proves the trust relationship works.
- **S3:** confirm flow log files are actually landing in the bucket.

9.6 Troubleshooting

Symptom	Likely cause	Fix
STS 400, ARN invalid	Missing `:role` in the ARN	Use the full form above
Credentials column blank	Credential not saved or selected	Re-add, select it, check Monitor status
HTML 500 error on adding the flow log	Cisco-side error	Reload and retry, confirm the path is listed, open a TAC case
Permissions tab token errors	Usually transient	Reload and retry; open a case if it persists
Nothing in Investigate or Assets	No traffic, wrong region, missing S3 read policy or credential	Check traffic, the policy on the role, and the credential

If the S3 buckets restrict access by IP, allow Cisco's published APJC source addresses (15 listed in XDR's integrations documentation).

9.7 Next steps once data flows

Check Investigate for AWS activity, review Assets for new devices, and turn on the AWS detections you need. XDR lists no response actions for AWS, so containment of cloud resources would need custom workflows that call AWS APIs. Start with notify and enrich only.

10. AI assistant and agent

The two AI rows in the sheet are "Use the AI assistant to interact with an incident" (marked Roadmap) and "AI Incident Summarization"; the agent creation form was worked through, but its use on an incident is not yet confirmed.

This section is the least certain in the report. The form's exact fields and the models offered differ by tenant and release, and the Target field was still open when the session ended. Confirm each field on your own screen.

10.1 Creating the agent

1. Open the AI assistant or agent area in XDR and choose to create a new agent.
2. **Name and description.** Name it for its job, for example `Incident Triage Assistant`, and describe it in one sentence.
3. **Instructions.** This is the most important field. Give it a role, the steps to follow, what it may and may not do, and the format of its answer. A starting point:

```
You help a SOC analyst triage Cisco XDR incidents.
For the incident you are given:
1. Summarise what happened in three sentences.
2. List the affected assets and users.
3. List each observable with its verdict.
4. Suggest the next three steps for the analyst.
Do not isolate hosts, block anything, or change the incident.
If data is missing, say so instead of guessing.
```

4. **Target or tools.** Select the target or workflows the agent may use. Give it read-only lookups first. Do not give it isolation until the instructions and results have been tested.
5. Save and test.

10.2 Using it in incident response

- Open an incident and ask the assistant for a summary and a timeline. This covers both AI rows.
- Ask it to explain an observable or suggest the next playbook task.
- Keep containment as a human-approved step, in line with the playbook (section 6.2).

10.3 Proof for the sheet

Capture a screenshot of the agent form, a screenshot of the assistant answering about a real incident, and a short note of the incident ID. Because the first row is marked Roadmap, state in the Notes column whether the feature was available in the tenant.

11. Testing and simulation

Test each feature in the order below, and only on machines you own, with the approval of whoever monitors the network.

11.1 Order of testing

1. Run the workflow by hand with a real observable.
2. Fire the automation rule with a loose condition, then tighten it.
3. Create a **new** incident so the assignment rule attaches the playbook.
4. Walk every playbook task and then read the worklog.

11.2 Producing an incident from the lab machine

Method	What it proves	Notes
EICAR test file (from eicar.org)	Secure Endpoint detects a harmless test file and reports it to XDR	Safe and standard
Manual workflow run with the hostname or GUID	The containment workflow, without waiting for a detection	Used to prove isolation
Temporarily lowered rule threshold	The trigger fires on any incident	Raise it back to 800 afterwards
Approved detection scenarios from Cisco or HCL security	Specific XDR analytics	Ask the Cisco contact which scenarios are approved for your tenant

The file `Cisco_XDR_Alert_Triggers.docx` lists alert names, the telemetry each one needs, and test commands. Some commands imitate real malware activity, so they are left out of this report on purpose. Read the file for its alert names and requirements, and let authorised staff run its commands in an isolated lab.

To use any of those alerts, the machine must be monitored by XDR, the alert must be enabled in XDR Analytics, and a lookback period can suppress repeat alerts.

11.3 Pre-flight checks

- Confirm each integration shows Connected before building on it.
- Confirm the workflow is Validated; a rule will not run an unvalidated workflow.
- Run `hello_xdr_api.py` to prove the API client and token work in your region.

11.4 After each test

Reverse containment on the lab machine, set the rule threshold back, and save run links and incident IDs for section 13.

12. Troubleshooting log

Every error met during the build, with the cause and fix. When a run fails, open its Runs entry and read the **child step** that failed.

Area	Symptom	Cause	Fix
Workflow	Set Variables invalid on validation	Variable not declared	Declare it as Input, Output or Static
Workflow	Install "succeeded with warnings"	Validation skipped	Resolve the warnings first
Workflow	Run "succeeds" but did nothing	Continue on failure is ticked	Untick it while testing
Workflow	Output shows `{}`	No output variables mapped	Map workflow outputs (section 7.4)
Rule	Workflow never starts	Not validated, not attached, or condition never matches	Validate, attach, and test with a loose condition
Playbook	Not on an incident	Assignment rules apply only at creation	Test with a new incident
Playbook	Workflow missing from task picker	Not validated or not enabled for incident response	Validate and check the workflow's settings
Webex	"No valid account key"	Integration not connected	Enable Webex in Integrations
Webex	"Execute on this target group"	No target chosen	Pick the Webex target in the workflow
Webex	`room ... not found`	Room name does not match a space the bot is in	Use the exact space name, or create it and add the bot
Webex	Placeholder text in the message	Variables not inserted	Replace each with the picker value
Webex	Two messages per incident	Webex in rule and playbook	Keep it in one place
Email	Target validation error	SMTP sign-in rejected	Verified sender, Gmail app password, or IT enables SMTP AUTH
ServiceNow	401	Bad credentials or reset required	Reset the password and untick Password needs reset
ServiceNow	403	Missing role	Add `itil`
ServiceNow	404 or timeout	Wrong host or sleeping instance	Check the host and wake the instance
Jira	Run fails	Wrong project key or issue type	Key without `-1`, a real issue type, clear ID and Watchers

Area	Symptom	Cause	Fix
Secure Endpoint	Run ends with no isolation	Observable type or GUID not recognised, or Host Isolation off	Run with the hostname, check the iteration, enable Host Isolation
AWS	STS 400 invalid ARN	Missing `:role`	Full ARN with `:role/cisco-xdr-integration`
AWS	Blank credentials column or dropdown	Credential not saved or selected	Re-add and select it
AWS	HTML 500 on adding flow log	Cisco-side error	Retry and open a TAC case
AWS	Permissions token errors	Usually transient	Reload and retry
AWS	No data in Investigate or Assets	No traffic, region, policy or credential	See section 9.6
API	401	Expired token or wrong region host	Request a new token from the right host
API	403	API client lacks the scope	Edit the scopes

13. Evidence checklist and open items

Change a row's Requirement Met cell only after its evidence exists, and put incident IDs and run links in the Notes column.

13.1 Evidence per row

Sheet row	Evidence to capture	Mark Met when
Create Ticket	Workflow run, the ServiceNow ticket, and a note on the incident	Ticket has the right fields and a rerun does not duplicate it
Complete Containment Workflow	Workflow run with the isolate step, the isolated machine, console view, and the reversal	Host isolated and restored through XDR
Create a Trigger	Rule configuration, the notification received, and the run history	The rule fires on its own at the agreed score
Create a Custom Playbook	Playbook definition and a completed incident walk-through	Every task completes and the worklog is full
Public cloud	Credentials page showing Connected, flows in Activities	Data from AWS visible in XDR
AI rows	Agent form and an answer about a real incident	Feature used on an incident

13.2 Open items

- [ ] Confirm AWS data appears: re-add the credential with the correct ARN, check S3 files, check the `AssumeRole` event, or ask Cisco about `ap-south-2`.
- [ ] Install and test the ServiceNow Create or Update workflow, and confirm a second run updates the same ticket.
- [ ] Add the ServiceNow task to the playbook and publish a new version.
- [ ] Replace `<<PLACEHOLDER>>` text in the Webex message with picker variables.
- [ ] Keep Webex in the rule or the playbook, not both.
- [ ] Return the rule to standalone and align its title, condition and description.
- [ ] Add the output variables to the containment workflow (section 7.4).
- [ ] Add the containment task to the playbook and walk a new incident through it.
- [ ] Confirm the Jira failure is fixed, or retire Jira.
- [ ] Fix email delivery, or rely on Webex.
- [ ] Confirm with HCL whether Secure Endpoint or CrowdStrike is expected, and what "ERD-M" meant.
- [ ] Finish the AI agent form and test it on an incident.
- [ ] Agree Level of Importance values with HCL, then fill Requirement Met with the evidence above.